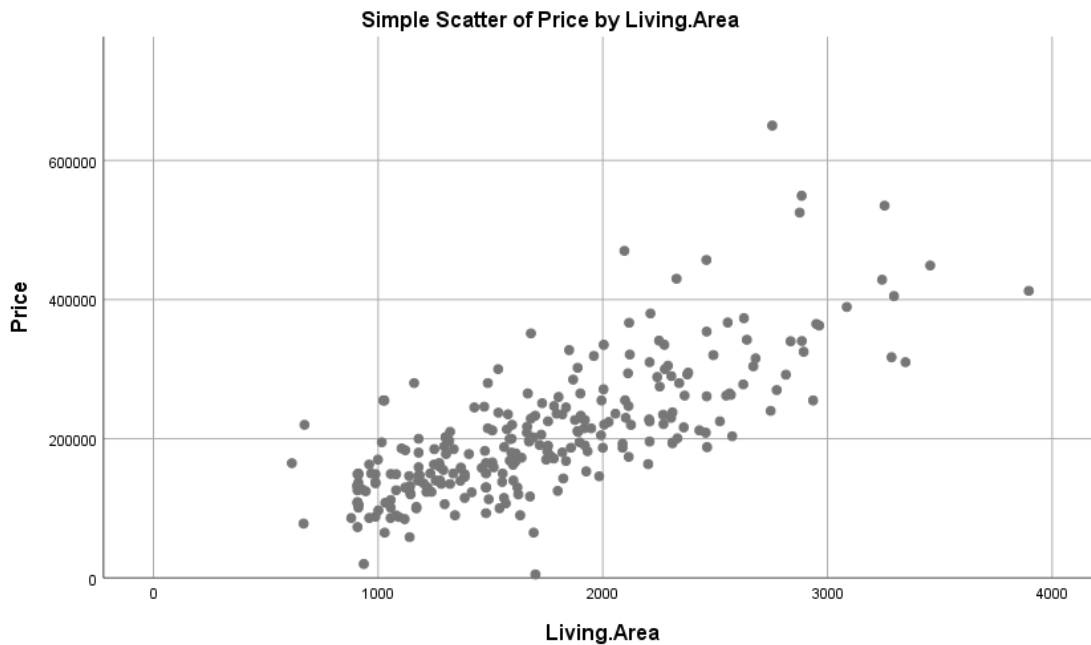


Construct a scatter plot of the selling price against the living area.



Regression analysis is a statistical tool for studying the relationship between 1 variable (response variable) and other variables (predictor variables). Predict the value of response variable based on the value(s) of predictor variable(s).

Selling price is selected to be the response variable is because, this project aims at examining how well 'living area', can predict 'selling price'.

A linear regression relation between the selling price against the living appears to be reasonable.

Regress the selling price on the living area

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	4794.275	11384.935		.421	.674
	Living.Area	116.652	6.182	.760	18.870	.000

a. Dependent Variable: Price

Estimated regression function:

$$\hat{Y} = 4794.275 + 116.652 * X$$

superimpose it on the scatter plot in part (1) as below:



According to the **ANOVA** table from SPSS, report Mean Square of Residual, **MSE** = 3791575129.191

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1350146028816.790	1	1350146028816.790	356.091	.000 ^b
	Residual	989601108718.938	261	3791575129.191		
	Total	2339747137535.728	262			

a. Dependent Variable: Price

b. Predictors: (Constant), Living.Area

According to the **Model Summary^b** table from SPSS, report **R²** = 57.7%.

R² Interpretation: A linear regression relation between the selling price against the living accounts for 57.7% of the total variation. The relationship between those two variables explains of 57.7% of the variation of the data, which is not that significant.

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.760 ^a	.577	.575	61575.767

a. Predictors: (Constant), Living.Area

b. Dependent Variable: Price

90% confidence interval for β_1 , can be reported from below **Coefficients^a** table from SPSS, it is Lower Bound 106.448 to Upper Bound 126.856. is be interpreted as the interval which covers 90% of the sample mean.

Coefficients^a								
		Unstandardized Coefficients		Standardized Coefficients			90.0% Confidence Interval for B	
Model		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound
1	(Constant)	4794.275	11384.935		.421	.674	-13998.982	23587.532
	Living.Area	116.652	6.182	.760	18.870	.000	106.448	126.856

a. Dependent Variable: Price

The alternatives:

$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

Decision rule:

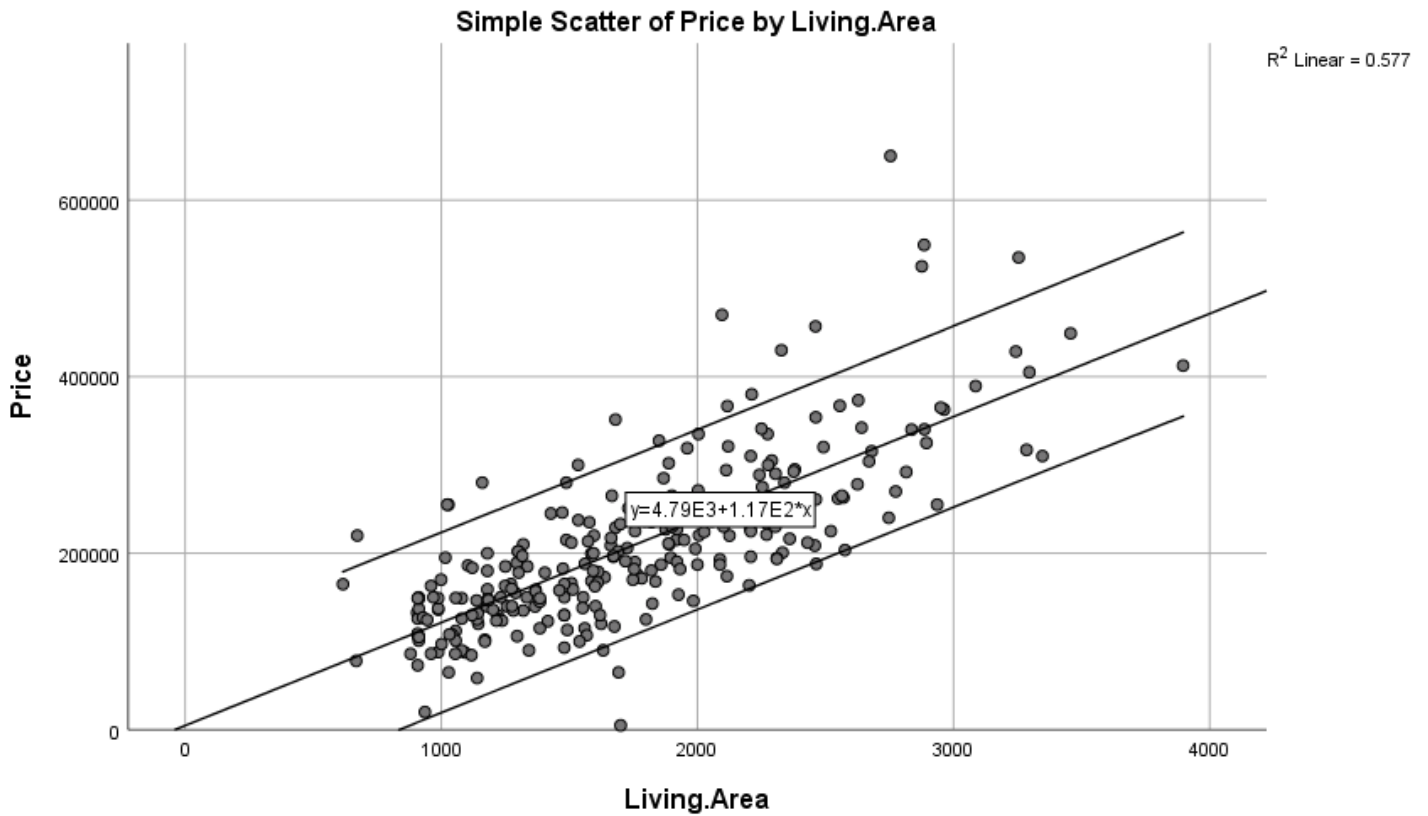
P-Value rule. Reject H_0 when P-Value is smaller than 0.10.

Conclusion:

For p-value = 0, which is smaller than 0.10, we reject H_0 , conclude H_a , that $\beta_1 \neq 0$.

Read the output value from SPSS output, that:

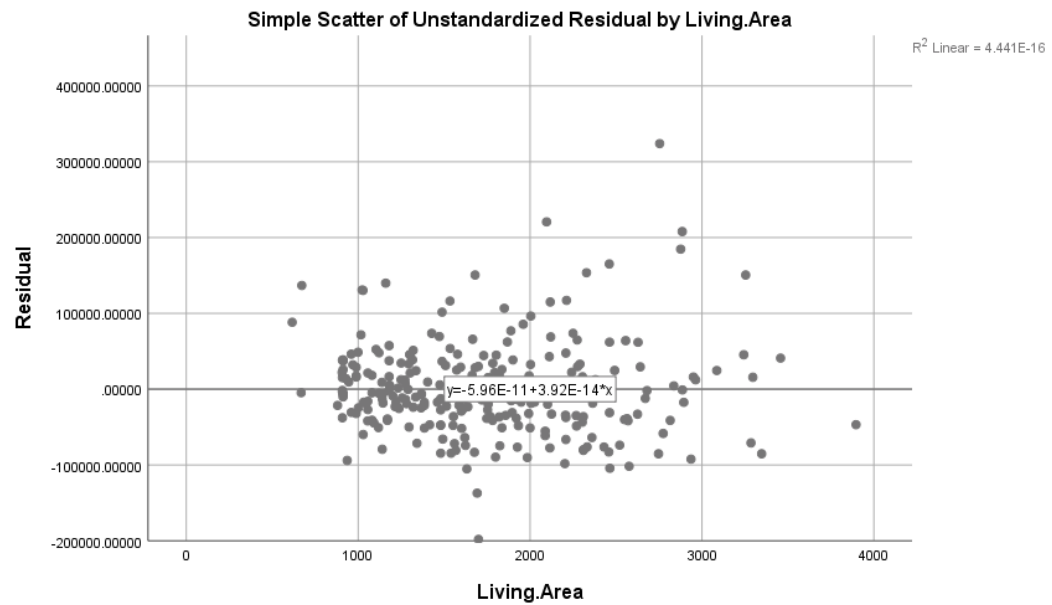
- The predicted selling price is about 220600 dollars, when a home with 1850 square feet (Residual is 106755).
- 90% prediction interval at this point is: from 118756.62460 dollars to 322443.76916 dollars



Model diagnostics on a residual plot against X.

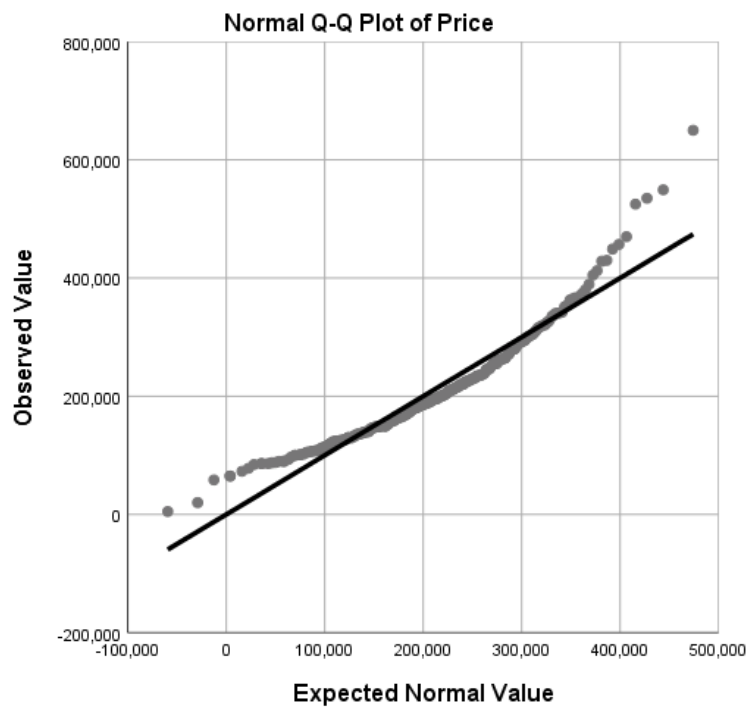
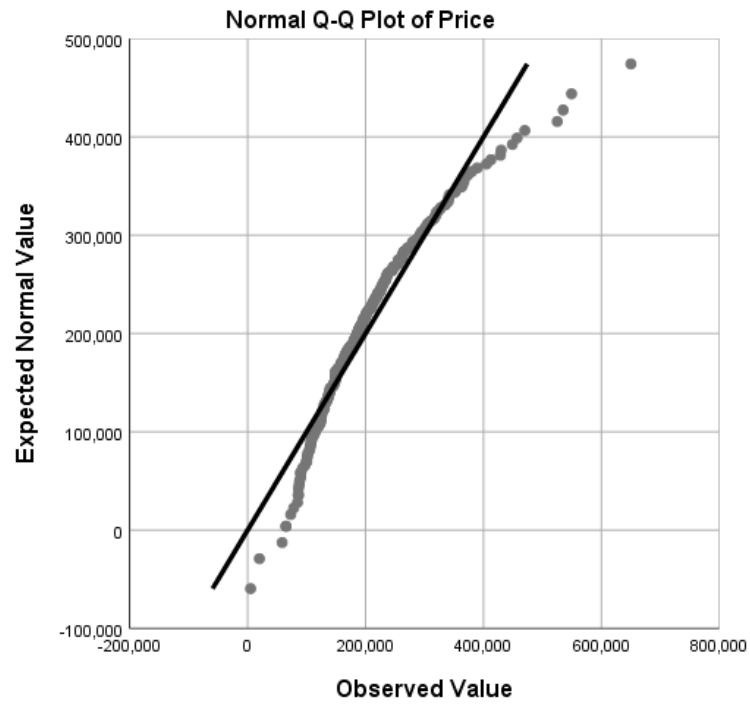
- Conclude no violation of linearity of the relationship between selling price and living area, for below plot shows good linearity, no strong evidence to conclude the violation.

- Conclude no violation of constant residual variance of the relationship between selling price and living area, for below plot shows constant variance, except couples of outliers might cause risk on violation of normality, no strong evidence to conclude the violation.



Model diagnostics on normal probability plots, Q-Q plots:

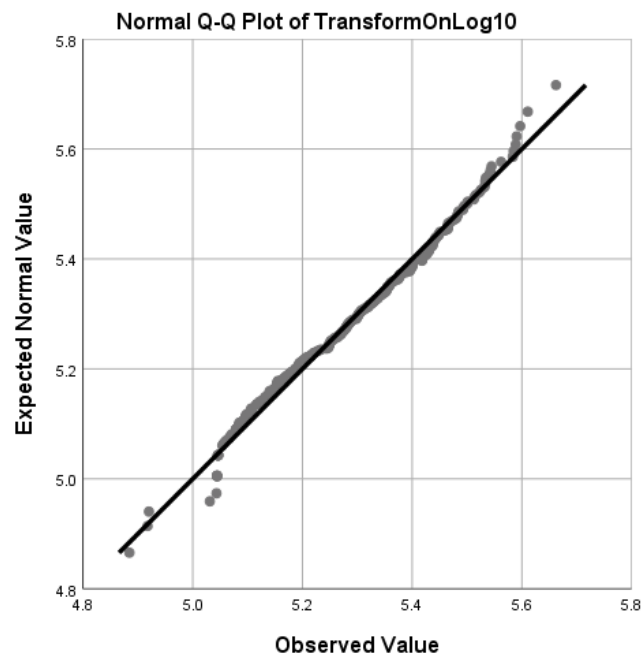
- Conclude violation of normality. Because the Q-Q plots are not showing nearly linear and indicate the heavy tail.



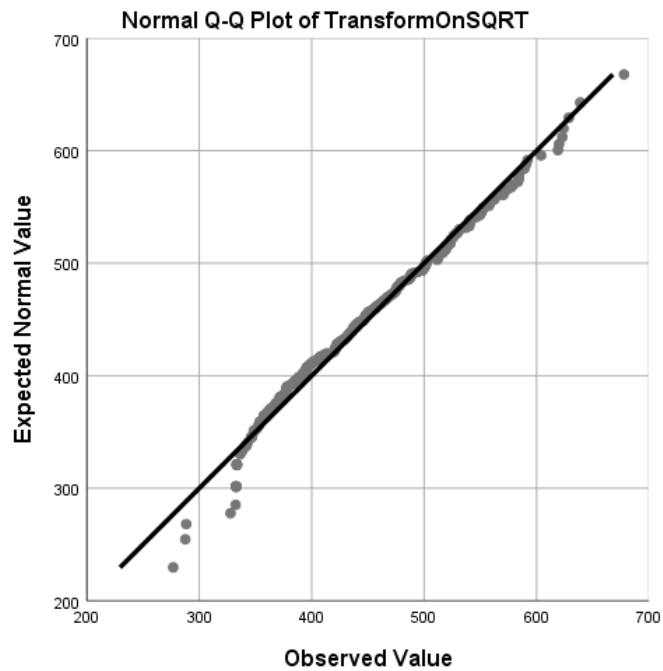
There is no violation of linearity, so no need to do transformations on X. while there is the violation of normality, which can be remedy by transformations of Y. And there are 3 potential ways of transformation might help, we will try one by one. And see which is the best way of remedy of normality.

1. $Y' = \log_{10} Y$
2. $Y' = \text{SQRT}(Y)$
3. $Y' = 1/Y$

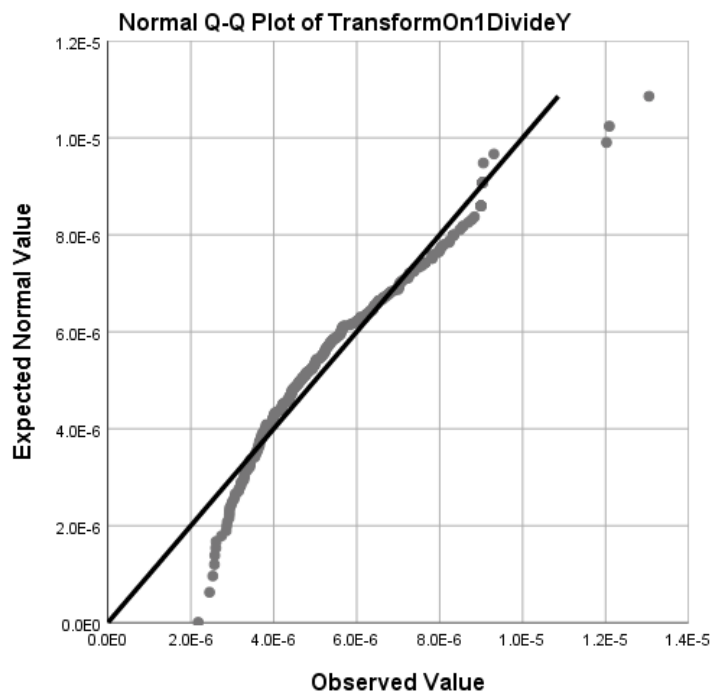
After transformation on Y, $Y' = \log_{10} Y$, there shows **better** normality. We can take this transformation as remedy of normality.



After transformation on Y , $Y' = \text{SQRT}(Y)$, there shows **better** normality. We can take this transformation as remedy of normality.



After transformation on Y , $Y' = 1/Y$, there shows **even worse** normality. So we reject this transformation as a remedy.



Model #1

The LS regression function

$$E\{Y\} = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3}$$

X_1 : living area

X_2 : number of bedrooms

X_3 : number of fireplaces

Using SPSS, regress Y on X_1 , X_2 and X_3 . Read result and the function is:

$$E\{Y\} = 28357.351 + 131.440 X_{i1} - 15873.874 X_{i2} + 1138.275 X_{i3}$$

Coefficients ^a								
		Unstandardized Coefficients		Standardized Coefficients			95.0% Confidence Interval for B	
Model		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound
1	(Constant)	28357.351	14661.995		1.934	.054	-514.544	57229.246
	Size of the living area of house in square feet	131.440	9.405	.856	13.975	.000	112.920	149.961
	Number of bedrooms	-15873.874	6355.692	-.141	-2.498	.013	-28389.283	-3358.464
	Number of fireplaces	1138.275	7961.437	.006	.143	.886	-14539.112	16815.662

a. Dependent Variable: Price of house as sold in 2002

$\beta_2 = -15873.874$, stands for $\Delta E\{Y\}$ will be - **15873.874**, when $\Delta X_2 = 1$, holding X_1 and X_3 to be constant. And the Negatives sign indicates the number of bedroom increase, the price will drop, when keep other parameter fixed. It is basically saying, when the total living area fixed and fixed number of fireplace, the more of bedroom indicates smaller area of each room, which indicates less price.

Using $\alpha = 0.05$, conduct an **F** test of overall linear relationship. State the alternatives, the value of F-test statistic, p-value, and your conclusion.

Model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \epsilon_i, \text{ where } \epsilon_i \sim \text{iid } N(0, \sigma^2) \text{ for } i = 1, 2, \dots, 263$$

Alternatives:

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0$$

H_a : at least one of them are not 0

Perform an F test using SSPS

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1373863789935.211	3	457954596645.071	122.800	.000 ^b
	Residual	965883347600.516	259	3729279334.365		
	Total	2339747137535.728	262			

a. Dependent Variable: Price of house as sold in 2002

b. Predictors: (Constant), Number of fireplaces, Number of bedrooms, Size of the living area of house in square feet

Read result from SSPS output:

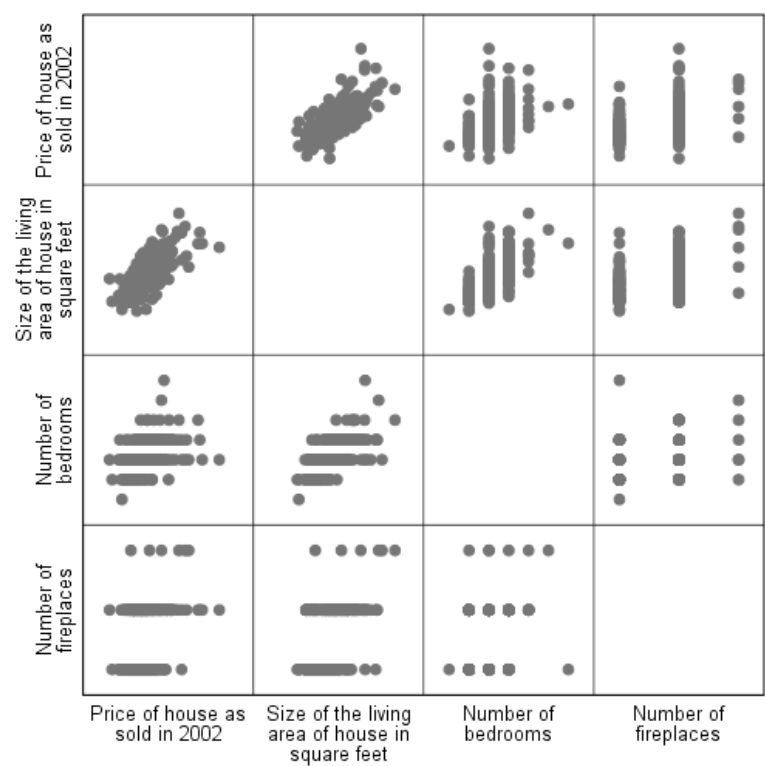
$$F^* = 122.800$$

$$p\text{-value} = 0.000$$

Conclusion:

Reject H_0 , because $p\text{-value} = 0.000 < \alpha = 0.05$. Indicates there is regression relationship exists.

Obtain the scatter plot matrix and the correlation matrix of the three predictor variables.



Correlations

		Price of house as sold in 2002	Size of the living area of house in square feet	Number of bedrooms	Number of fireplaces
Price of house as sold in 2002	Pearson Correlation	1	.760**	.464**	.365**
	Sig. (2-tailed)		.000	.000	.000
	N	263	263	263	263
Size of the living area of house in square feet	Pearson Correlation	.760**	1	.704**	.464**
	Sig. (2-tailed)	.000		.000	.000
	N	263	263	263	263
Number of bedrooms	Pearson Correlation	.464**	.704**	1	.275**
	Sig. (2-tailed)	.000	.000		.000
	N	263	263	263	263
Number of fireplaces	Pearson Correlation	.365**	.464**	.275**	1
	Sig. (2-tailed)	.000	.000	.000	
	N	263	263	263	263

** . Correlation is significant at the 0.01 level (2-tailed).

We can clearly see the collinearity among predictor variable **Number of bedrooms** and **Size of the living area of house in square feet**. And the Correlations between them is **0.704**, which is a relatively high correlation.

Compute Type I Sum of Squares using SSPS, Read from output:

$$SSR(X_3 | X_1, X_2) = 76231874.901$$

Tests of Between-Subjects Effects

Dependent Variable: Price of house as sold in 2002

Source	Type I Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	1373863789935.212 ^a	3	457954596645.071	122.800	.000
Intercept	11305518304631.234	1	11305518304631.234	3031.556	.000
Living.Area	1350146028816.793	1	1350146028816.793	362.039	.000
Bedrooms	23641529243.521	1	23641529243.521	6.339	.012
Fireplaces	76231874.901	1	76231874.901	.020	.886
Error	965883347600.515	259	3729279334.365		
Total	13645265442167.000	263			
Corrected Total	2339747137535.727	262			

a. R Squared = .587 (Adjusted R Squared = .582)

Compute the coefficient of partial determination and the coefficient of partial correlation, respectively, between Y and X3 given that X1 and X2 are in a model.

Read the output value from SSPS output, between Y and X3 given that X1 and X2 are in a model that:

- the coefficient of partial determination:

$$SSR(X_3 | X_1, X_2) = 76231874.901$$

- the coefficient of partial correlation

$$(SSR(X_3 | X_1, X_2))^{1/2} = 8731.087$$

Using $\alpha = 0.05$, conduct an F test of whether X_3 is helpful, given that X_1 and X_2 are in a model. State the alternatives, the value of F-test statistic, p-value, and your conclusion.

Alternatives:

$$H_0: \beta_3 = 0$$

$$H_a: \beta_3 \neq 0$$

Perform an F test using SSPS compute Type III Sum of Squares using SSPS,

Tests of Between-Subjects Effects

Dependent Variable: Price of house as sold in 2002

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	1373863789935.212 ^a	3	457954596645.071	122.800	.000
Intercept	13949868927.854	1	13949868927.854	3.741	.054
Living.Area	728328703509.720	1	728328703509.720	195.300	.000
Bedrooms	23262964108.628	1	23262964108.628	6.238	.013
Fireplaces	76231874.901	1	76231874.901	.020	.886
Error	965883347600.515	259	3729279334.365		
Total	13645265442167.000	263			
Corrected Total	2339747137535.727	262			

a. R Squared = .587 (Adjusted R Squared = .582)

Read result from SSPS output:

$$F^* = 0.020$$

$$p\text{-value} = 0.886$$

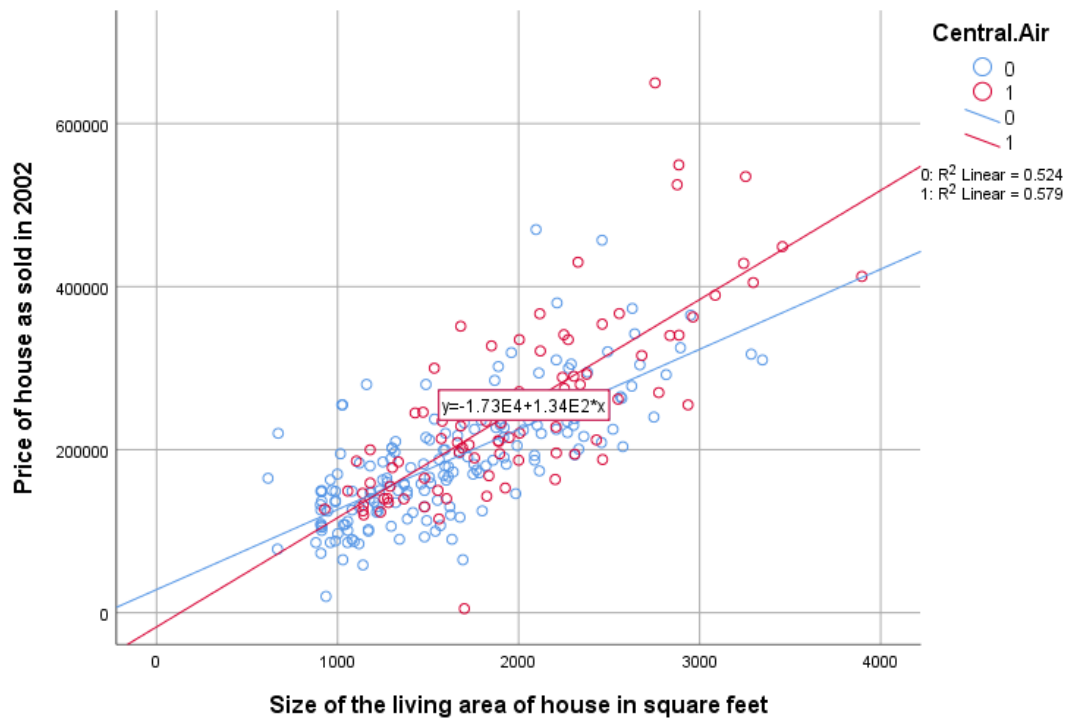
Conclusion:

Falls to reject H_0 , because $p\text{-value} = 0.886 > \alpha = 0.05$. Indicates X_3 is helpful, given that X_1 and X_2 are in a model.

Model #2:

Construct a scatterplot of the selling price against the living area for the two groups:

Central Air = 1 and = 0.



Because the two fitted line for each group are not parallel to each other. Indicating that they are not independent to each other, so we should include the interaction term.

The LS regression function and interpret the coefficients of Central Air and the interaction term, respectively.

$$E\{Y\} = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_1 X_2$$

X_1 : living area

X_2 : Central Air (0/1)

$X_1 X_2$: Interaction

Using SPSS, regress Y on X_1 , X_2 and $X_1 X_2$. Read result and the function is:

$$E\{Y\} = 28420.679 + 98.245X_{i1} - 45769.889X_{i2} + 35.606 X_{i1} X_{i2}$$

β_2 : When living area is fixed, if Central Air is 1, indicated decrease of 45769.889 of the Price. Otherwise no contribution the price.

β_3 : When Central Air is 1, one unit of increase of living area indicated increase of 35.606 of the Price. Otherwise no contribution the price.

Coefficients ^a							
		Unstandardized Coefficients		Standardized Coefficients		95.0% Confidence Interval for B	
Model		B	Std. Error	Beta	t	Sig.	
							Lower Bound Upper Bound
1	(Constant)	28420.679	13801.434		2.059	.040	1243.371 55597.987
	Size of the living area of house in square feet	98.245	8.109	.640	12.115	.000	82.277 114.214
	Central.Air	-45769.889	25468.621	-.228	-1.797	.073	- 4382.043
	InteractionTerm	35.606	13.024	.380	2.734	.007	9.960 61.252

a. Dependent Variable: Price of house as sold in 2002

Using $\alpha = 0.05$, conduct a partial test for identical regression function.

Alternatives:

$H_0: \beta_1 = \beta_2 = \beta_3 = 0$

H_a : at least one of them are not 0

Perform an F test using SSPS